

Fourier integrals (11.7)

Ex

Fourier integrals can be written as

$$f(x) = \int_0^{\infty} [A(\omega) \cos \omega x + B(\omega) \sin \omega x] d\omega \quad (1)$$

where

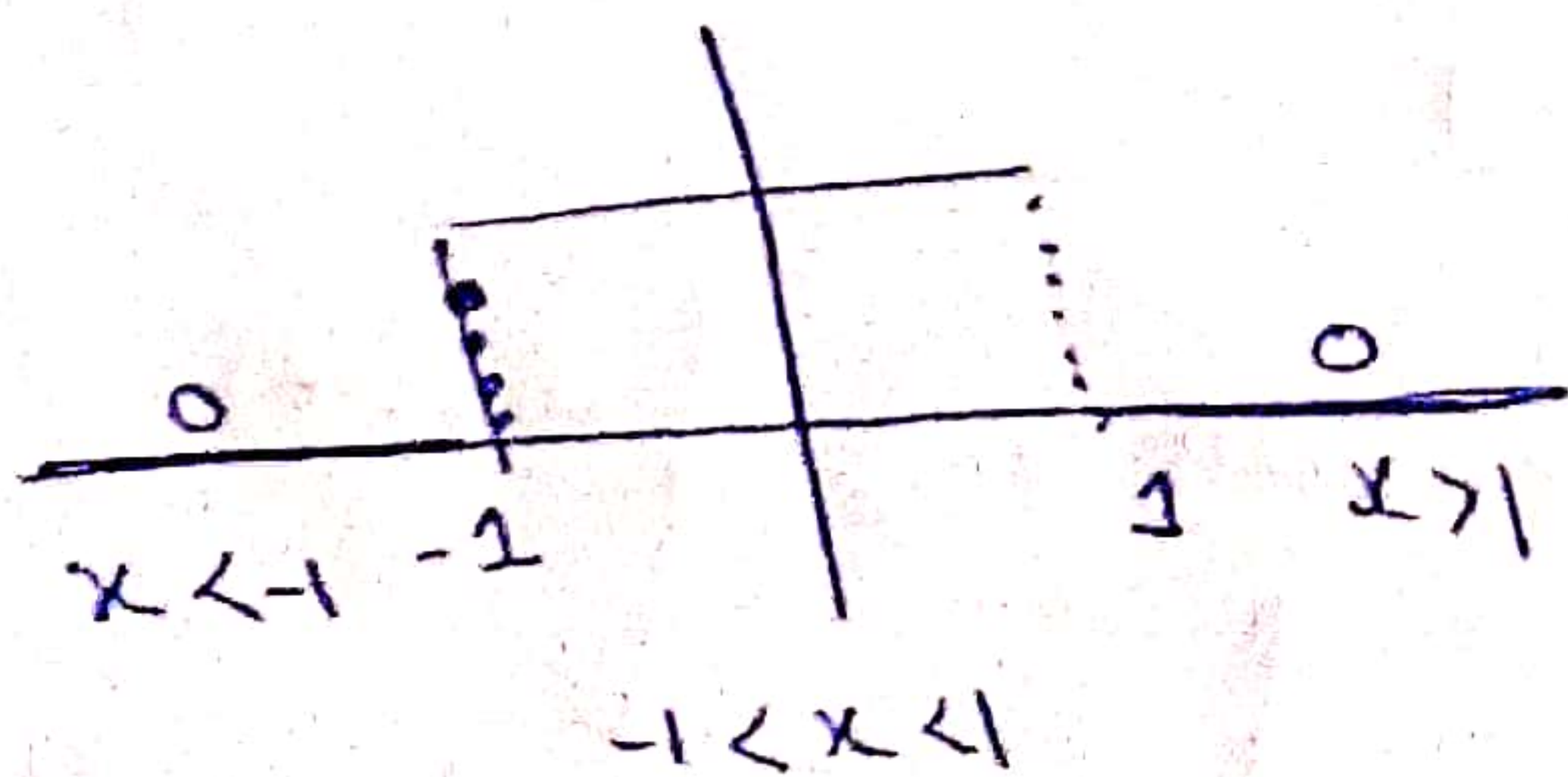
$$A(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(\eta) \cos \omega \eta d\eta \quad (2)$$

$$B(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(\eta) \sin \omega \eta d\eta \quad (3)$$

Ex

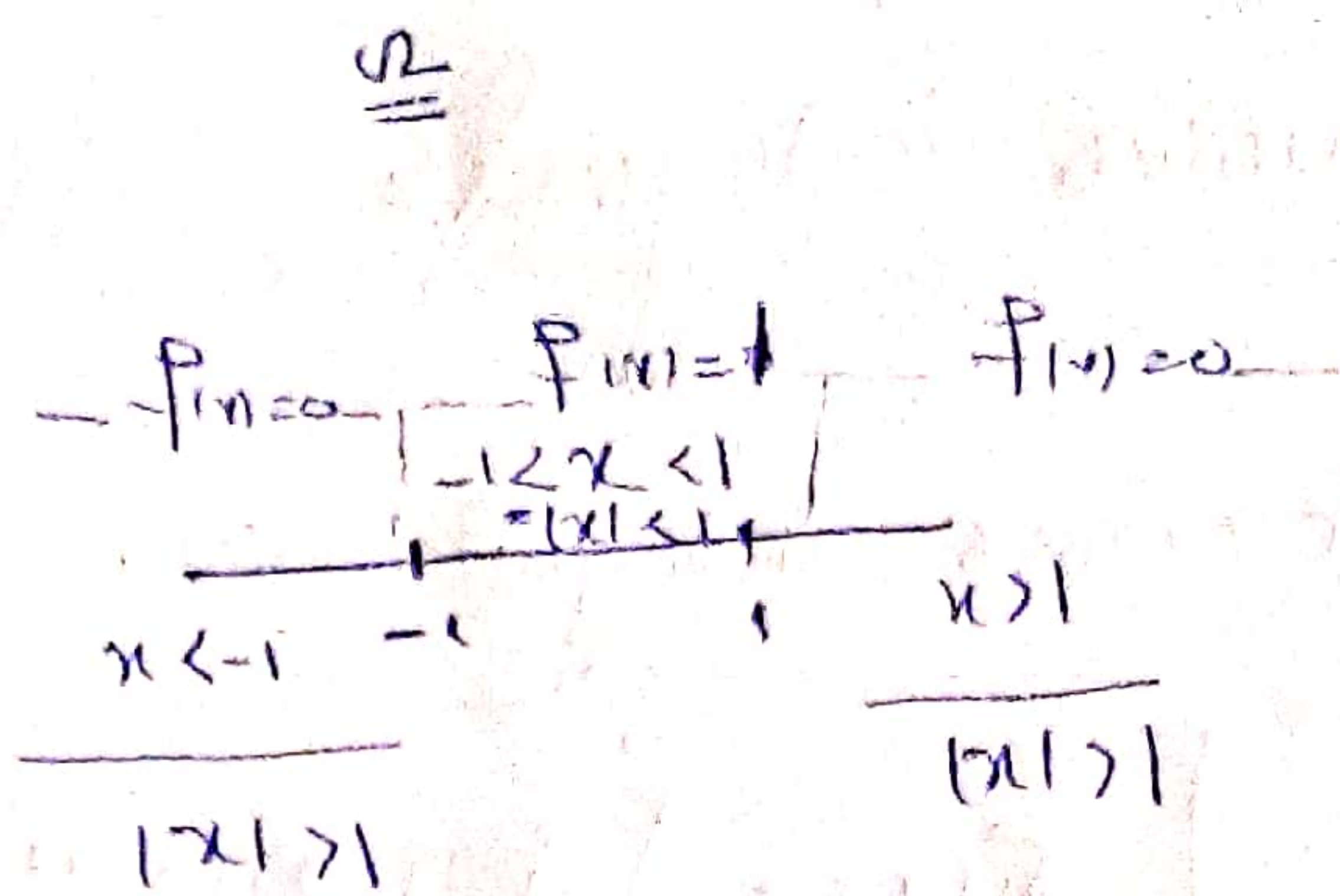
Find Fourier integral representation of the following function.

$$f(x) = \begin{cases} 1 & \text{if } |x| < 1 \\ 0 & \text{if } |x| > 1 \end{cases} \quad (*)$$



Sol

$$A(\omega) = \int_{-\infty}^{\infty} f(\eta) \cos \omega \eta \, d\eta$$



$$= \int_{-\infty}^{-1} \cancel{f(\eta)} \cos \omega \eta \, d\eta + \int_{-1}^1 f(\eta) \cos \omega \eta \, d\eta + \int_1^{\infty} \cancel{f(\eta)} \cos \omega \eta \, d\eta$$

$$= \int_{-1}^1 \cos \omega \eta \, d\eta = \left. \frac{\sin \omega \eta}{\omega} \right|_{-1}^1 = \frac{1}{\omega} [\sin \omega(1) - \sin \omega(-1)] = \frac{2 \sin \omega}{\omega}$$

$$B(\omega) = \int_{-\infty}^{\infty} f(\eta) \sin \omega \eta \, d\eta$$

$$= \int_{-\infty}^{-1} \cancel{f(\eta)} \sin \omega \eta \, d\eta + \int_{-1}^1 f(\eta) \sin \omega \eta \, d\eta + \int_1^{\infty} \cancel{f(\eta)} \sin \omega \eta \, d\eta$$

$$= \int_{-1}^1 \sin \omega \eta \, d\eta = -\frac{\cos \omega \eta}{\omega} = -\frac{1}{\omega} [\cos(\omega) - \cos(-\omega)] = 0$$

As,

$$f(x) = \int_0^{\infty} \left(A(\omega) \cos \omega x + B(\omega) \sin \omega x \right) d\omega$$

So,

$$f(x) = \int_0^{\infty} 2 \frac{\sin \omega}{\omega} \cos \omega x d\omega$$

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\sin \omega \cos \omega x}{\omega} d\omega$$

which is the generalized representation

Exercise

show that

following

$$\text{Q1} \quad \int_0^{\infty} \frac{\cos x\omega + \omega \sin x\omega}{1 + \omega^2} d\omega = \pi e^{-x} \quad \text{if } x > 0$$

$$\text{Q2} \quad \int_0^{\infty} \frac{\omega^3 \sin x\omega}{\omega^4 + 4} d\omega = \frac{1}{2} \pi e^{-x} \quad \text{if } x > 0$$

$$\text{Q3} \quad \int_0^{\infty} \frac{\sin \omega x - \omega \cos \omega x}{\omega^2} d\omega = \frac{\pi x}{2} \quad \text{if } 0 < x < 1$$

11.9 Fourier Transform

LPO

(Discrete and Fast Fourier Transform)

we have for a function $f(x)$, if complex
Fourier transform is given as,

$$F(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx \quad (1)$$

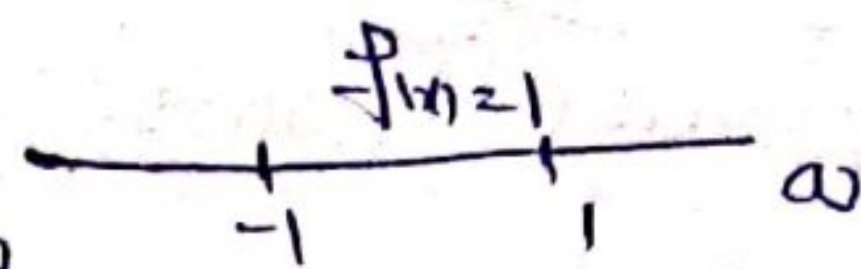
and its inverse Fourier transform is

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(\omega) e^{i\omega x} d\omega$$

(2)

Ex #01

Find Fourier transform of $f(x)$



$$f(x) = \begin{cases} 1 & -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$



Sol:

$$F(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx \quad (1)$$

$$F(\omega) = \frac{1}{\sqrt{2\pi}} \left[\int_{-1}^1 f(x) e^{-i\omega x} dx + \int_{-1}^1 f(x) e^{-i\omega x} dx \right] \quad (2)$$

$$F(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-1}^1 e^{-i\omega x} dx \quad (3)$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-i\omega x}}{-i\omega} \right]_{-1}^1$$

$$= \frac{1}{\sqrt{2\pi} (-i\omega)} [e^{-i\omega} - e^{i\omega}] \quad (4)$$

(Q2)

$$F(\omega) = \frac{1}{\sqrt{2\pi}(-i\omega)} \left[\cancel{\cos \omega} - \cancel{2i \sin \omega} - \cancel{\cos \omega} - \cancel{2i \sin \omega} \right]$$

$$= \frac{-2i \sin \omega}{\sqrt{2\pi}(-i\omega)}$$

$$= \frac{2}{\sqrt{2\pi}} \frac{\sin \omega}{\omega}$$

Ex #02

Find Fourier transform of

$$f(x) = \begin{cases} x & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases}$$

Sol

$$F(\omega) = \frac{1}{\sqrt{2\pi}} \left[\int_{-\infty}^0 f(x) e^{-i\omega x} dx + \int_0^a f(x) e^{-i\omega x} dx \right]$$

$$F(\omega) = \frac{1}{\sqrt{2\pi}} \left[\int_0^a x e^{-i\omega x} dx \right]$$

$$= \frac{1}{\sqrt{2\pi}} \left[x \frac{e^{-i\omega x}}{-i\omega} \right]_0^a + \int_0^a \frac{e^{-i\omega x}}{i\omega} dx$$

$$= \frac{1}{\sqrt{2\pi}} \left[-\frac{1}{i\omega} (a e^{-i\omega a}) + \frac{1}{i\omega} \frac{e^{-i\omega x}}{-i\omega} \right]_0^a$$

$$= \frac{1}{\sqrt{2\pi}} \left[-\frac{1}{i\omega} (a e^{-i\omega a}) + \frac{1}{\omega^2} (e^{-i\omega a} - 1) \right]$$

= simplify further —

Exercise

Q1 $f(x) = \begin{cases} x e^{-x} & \text{if } -1 < x < 0 \\ 0 & \text{otherwise} \end{cases}$

Q2 $f(x) = \begin{cases} e^{xx} & \text{if } x < 0 \\ 0 & \text{if } x > 0 \end{cases}$